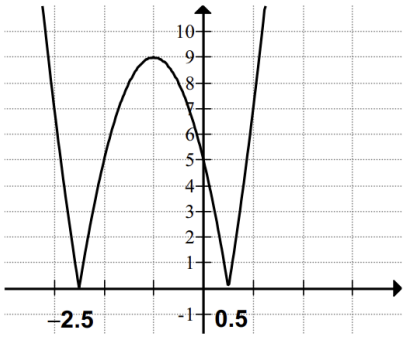
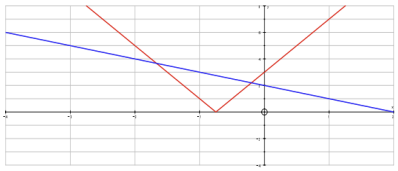
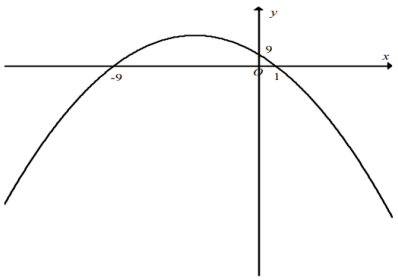
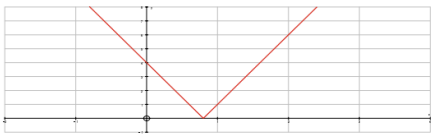
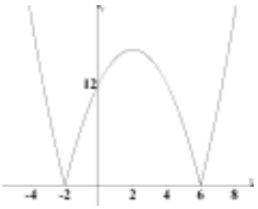
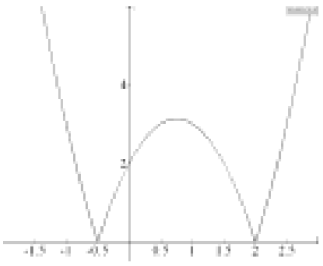
Additional Mathematics (0606)

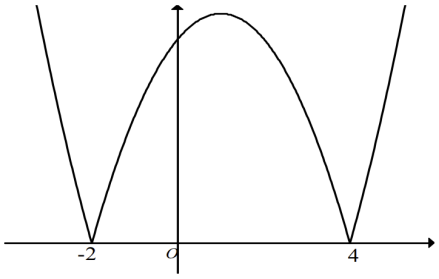
Mark Scheme Paper 2 (Practice) — **Non-Calculator**
Chapter 1: Functions — **Sources:** CIE 0606 Past Papers
Compiler: Kimhan — **Quality Control**: Kimhan

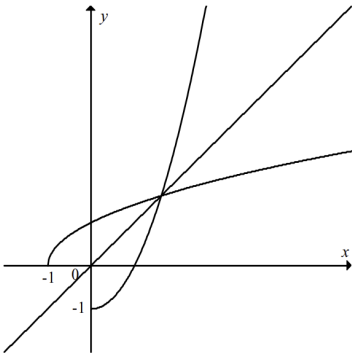
Question	Answer	Marks	Partial Marks
1	**(i)** $2-\sqrt{5}<\mathrm{f}(x)\leqslant 2$	**B2**	**B1** for $\leqslant 2$ **B1** for $2-\sqrt{5}<$ or awrt –0.24 Must be using f, f(x) or y, $2-\sqrt{5}<$, if not then **B1** max
	(ii) $\mathrm{f}^{-1}(x)=(2-x)^2-5$ Domain $2-\sqrt{5}<x\leqslant 2$ Range y or $-5\leqslant \mathrm{f}^{-1}(x)<0$	**M1** **A1** **B1** **B1**	for a correct method to find the inverse Must be using the correct variables for the B marks
	(iii) $\mathrm{fg}(x)=\mathrm{f}\left(\frac{4}{x}\right)$ $=2-\sqrt{\frac{4}{x}+5}$ leading to $x=-4$	**M1** **DM1** **A1**	for correct order of functions for solution of equation
2	**(i)** $4(x+1)^2-9$	**B3,2, 1,0**	one mark for each of p, q, r correct in a correctly formatted expression; allow correct equivalent values; If **B0** then **SC2** for $4(x+1)-9$ or **SC1** for correct 3 values seen in incorrect format e.g. $4(x+1x)-9$ or $4(x^2+1)-9$ or for a correct completed square form of the original expression in a different but correct format. e.g. $2(\sqrt{2}x+\sqrt{2})^2-9$

	(ii) $(-1, 9)$ **(iii)** 	**B2FT** **B1** **B1** **B1**	**B1FT** $(-q\ ,-r)$ $r<0$ for each correct coordinate Correct symmetric W shape with cusps on x-axis y-intercept marked at 5 only or coords indicated on graph x-intercepts marked at –2.5 and 0.5 only x-axis or coords indicated on graph or close by
3	**(a)** Valid method when $x = \frac{1}{2}$ [greatest value =] $\frac{1}{4}$ **(b)** Valid comment e.g. when $x \geq 1$, f′ is always decreasing	**M1** **A1** **B1** **B1**	Completing the square as far as e.g. constant $-\left(x-\frac{1}{2}\right)^2$ or calculus as far as $1 - 2x = 0$ or finding roots $x = 0$ and $x = 1$and using symmetry soi Implies **M1** if not clearly from wrong working Allow e.g. a sketch with a comment such as the curve is one-one [when $x \geq 1$] or e.g. the curve is one-one when $x > \frac{1}{2}$
4	**(i)**  **(ii)** $2-x=3+2x$ leading to $x=-\frac{1}{3}$ $2-x=-3-2x$ leading to $x=-5$ Alternative: $(2-x)^2=(3+4x)^2$ leading to $15x^2+28x+5=0$ $x=-\frac{1}{3},\ \ x=-5$	**B1** **B1** **B1** **B1** **B1** **M1** **A1** **M1** **A1,A1**	for a line in correct position for $(0, 2)$, $(2, 0)$ for correct shape for $y=\lvert 3+2x\rvert$, touching the x-axis for $(-1.5, 0)$, $(0, 3)$ for $x=-\frac{1}{3}$ for correct attempt to deal with 'negative' branch. for $x=-5$ for equating and squaring to obtain a 3 term quadratic equation **A1** for each.

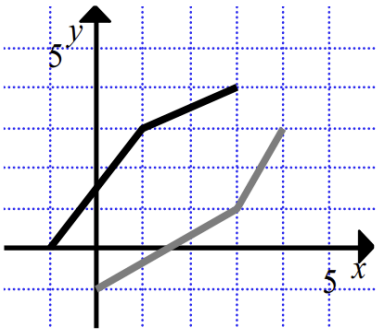
5	$4x-3=x \rightarrow x=1$ $4x-3=-x$ $x=0.6$ **OR** $(4x-3)^2=x^2$ $15x^2-24x+9=0$ $3(x-1)(5x-3)=0$ $x=1$ and $x=0.6$	**B1** **M1** **A1** **B1** **M1** **A1**	www use of $-x$ or $-(4x-3)$ but not both. solve correct 3 term quadratic www
6	**(i)** $\text{fg}(x)=\ln(2e^x+3)+2$ **(ii)** $\text{ff}(x)=\ln(\ln x+2)+2$	**B1** **B1**	isw isw (ignore subsequent working)
7	$[\text{f}(x)=]2x-4$ and $[\text{f}(x)=]-2x+4$	**B1,B1**	Condone y =….
8	**(a)** $(4x-1)(x+5)$ $[\leqslant 0]$ critical values $\frac{1}{4}$ and -5 soi $-5 \leqslant x \leqslant \frac{1}{4}$ **(b) (i)** $(x+4)^2-25$ or $a=4$ and $b=-25$ **(ii)** (Greatest value =) 25 $x=-4$ **(iii)** 	M1 A1 A1 B1, B1 B1ft B1ft B1 B1	Solves quadratic Accept: $\left[-5, \frac{1}{4}\right]$; $-5 \le x$ AND $x \le 0.25$ Must be clear Correct shape with maximum in second quadrant and crossing positive and negative axes correctly All 3 intercepts correctly shown on graph
9	**(i)**  **(ii)** $4-5x=\pm 9$ or $(4-5x)^2=81$ leading to $x=-1,\ x=\frac{13}{5}$	**B1** **B1** **B1** **M1** **A1, A1**	for shape, must touch the x-axis in the correct quadrant for y intercept for x intercept for attempt to obtain 2 solutions, must be a complete method **A1** for each

10	**(a)** **(i)** $(x+3)^2-5$ **(ii)** $y \geqslant 4$ or $f \geqslant 4$ **(iii)** $y=\sqrt{x+5}-3$ Domain $x \geqslant 4$	B1, B1 B1 M1 A1 B1FT	B1 for 3, B1 for -5 Correct notation or statement must be used for a correct attempt to find the inverse function must be in the correct form and positive root only Follow through on *their* answer to **(ii),** must be using x
11	**(i)**  **(ii)** $(2,16)$ $k=0$ **(iii)** $k>16$	B1 B1 B1 M1 A1 B1 B1	shape intercepts on x-axis (-2,0),(6,0) intercept on y-axis for a curve with a maximum and two arms (0,12) $(2, \pm 16)$ seen or $(2, k)$ where $k>0$ (2, 16) or $x=2$ **and** $y=16$ only
12	**(i)**  Maximum point occurs when $y=\frac{25}{8}$ **(ii)** so $k>\frac{25}{8}$	**B1** **B1** **B1** **M1** **A1**	**B1** for shape **B1** for $y=2$ (must have a graph) **B1** for $x=-0.5$ and 2 (must have a graph) **M1** for obtaining the value of y at the maximum point, by either completing the square, differentiation, use of discriminant or symmetry. Must have the correct sign for **A1** Ignore any upper limits

13	**(i)**  **(ii)** $x = 1$ (only) soi $y = \pm 9$ (only) $0 < k < 9$	**B1** **B1** **B1** **B1** **B1**	correct shape x intercepts marked or implied by tick marks, for example or seen nearby; condone y intercept omitted can be implied by second **B1** or $k = \pm 9, +9$ or -9 or both; must be strict inequality in k; condone space, comma, "and", "or"
14	**(i)** $2\left(x-\frac{1}{4}\right)^2+\frac{47}{8}$ (5.875) isw **(ii)** $\frac{47}{8}$ is min value when $x=\frac{1}{4}$	**B3,2,1,0** **B1ft + B1ft**	one mark for each of p, q, r correct; allow correct equivalent values. If **B0**, then **SC2** for $2\left(x-\frac{1}{4}\right)+\frac{47}{8}$, or **SC1** for correct values but incorrect format strict **ft** *their* $\frac{47}{8}$ and *their* $\frac{1}{4}$; each value must be correctly attributed; condone $y=\frac{47}{8}$ for **B1**, or $\left(\frac{1}{4},\frac{47}{8}\right)$ for **B1B1**
15	**(i)** f(3) $\frac{6}{4}$ oe **(ii)** $\dfrac{2\left(\frac{2x}{x+1}\right)}{\frac{2x}{x+1}+1}$ A correct and valid step in simplification Correctly simplified to $\frac{4x}{3x+1}$	**M1** **A1** **M1** **dM1** **A1**	or $fg(x)=\dfrac{2\sqrt{(x+1)}}{\sqrt{(x+1)}+1}$ allow omission of 2(…..) in numerator or (…..) + 1 in denominator, but not both. e.g. multiplying numerator and denominator by $x+1$, , or simplifying $\frac{2x}{x+1}+1$ to $\frac{2x+x+1}{x+1}$

	(iii) Putting $y = g(x)$, changing subject to x and swopping x and y or vice versa $g^{-1}(x) = x^2 - 1$ (Domain) $x > 0$ (Range) $g^{-1}(x) > -1$ **(iv)** 	**M1** **A1** **B1** **B1** **B1 + B1** **B1**	condone $x = y^2 - 1$; reasonable attempt at correct method condone $y = \ldots\ldots$, $f^{-1} = \ldots\ldots$ condone $y > -1$ $f^{-1} > -1$ correct graphs; -1 need not be labelled but could be implied by 'one square' idea of reflection or symmetry in line $y = x$ must be stated.
16	**(i)** $12\left(x - \frac{1}{4}\right)^2 + \frac{17}{4}$ isw **(ii)** *their* $\frac{4}{17}$ or *their* 0.235 *their* $x = \frac{1}{4}$ oe	**B3, 2, 1,0** **B1ft** **B1ft**	one mark for each of p, q, r correct in a correctly formatted expression; allow correct equivalent values; If **B0** then **SC2** for $12\left(x - \frac{1}{4}\right) + \frac{17}{4}$ or **SC1** for correct 3 values seen in incorrect format e.g. $12\left(x - \frac{1}{4}x\right) + \frac{17}{4}$ or $12\left(x^2 - \frac{1}{4}\right) + \frac{17}{4}$ or for a correct completed square form of the original expression in a different but correct format. e.g. $3\left(2x - \frac{1}{2}\right)^2 + \frac{17}{4}$ strict **ft** ; *their* $\frac{4}{17}$ must be a proper fraction or decimal rounded to 3sig figs or more or truncated to 4 figs or more strict **ft** ; x must be correctly attributed

17	**(i)** $3 < f < 7$	**B1,B1**	If **B0** then **SC1** for $3 < f < 7$
	(ii) $f(12) = 5$	**B1**	$f^2(x)$ $\sqrt{\left(\sqrt{(x-3)}+2-3\right)}+2$ earns **B1**
	$(f(5) =)\ 2+\sqrt{2}$	**B1**	
	(iii) Clear indication of method $f^{-1}(x) = (x-2)^2 + 3$	**M1** **A1**	condone $y = (x-2)^2 + 3$
	(iv) $gf(x) = \dfrac{120}{\sqrt{(x-3)}+2}$	**B1**	
	Attempt to solve *their* $gf(x) = 20$	**M1**	
	$x = 19$	**A1**	
18	**(i)** Idea of modulus correct	**B1**	Two straight lines above and touching x-axis
	$\frac{1}{2}$ indicated on x-axis	**B1**	Must be a sketch
	2 indicated on y-axis	**B1**	Must be a sketch
	(ii) $\frac{2}{3}$ (0.667)	**B1**	0.67 is **B0**
	Solve $4x - 2 = -x$ or $(4x-2)^2 = x^2$	**M1**	As far as x = numerical value
	$\frac{2}{5}$	**A1**	SC: If drawn then **B1**, **B2** for exact answers only

19	**(i)** 1	**B1**	Not a range for k, but condone $x = 1$ and $x \geqslant 1$
	(ii) $f \geqslant -5$	**B1**	Not x, but condone y
	(iii) Method of inverse	**M1**	Do not reward poor algebra but allow slips
	$1+\sqrt{x+5}$	**A1**	Must be $f^{-1} = ...$or $y = ...$
	(iv) f: Positive quadratic curve correct range and domain	**B1**	Must cross x-axis
	f^{-1}: Reflection of f in $y = x$	**B1√**	√*their* f*(x)* sketch Condone slight inaccuracies unless clear contradiction.
	(v) Arrange $f(x) = x$ or $f^{-1}(x) = x$ to 3 term quadratic = 0	**M1**	
	4 only www	**A1**	Allow $x = 4$ with no working. Condone (4, 4). Do not allow final **A** mark if –1 also given in answer
20	**(a)** **(i)** A and E	**B2**	1 mark for each **B1** for 1 extra, **B0** if 2 or more extras
	(ii) C and D	**B2**	1 mark for each **B1** if 1 extra, **B0** if 2 or more extras
	(b) 	**B2**	(–1, 0), (1, 3), (3, 4) or **B1** for two points correct and joined or for three points correct but clearly not joined

End of This Paper
Total Marks: 124